Geometric Intelligence: A Differential-Geometric Decision-Making Framework on Data-Driven Differentiable Manifolds

— Full Version —

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Abstract

We introduce **Geometric Intelligence (GI) theory**, a principled mathematical framework for constructing corporate management environments and national policy environments as data-driven Riemannian manifolds via variational autoencoders (VAEs), and for performing the full suite of differential-geometric computations—Riemannian metric evaluation, curvature tensor analysis, Lie derivative assessment, geodesic computation, and optimal control simulation—on these manifolds so as to generate actionable intelligence for strategic decision-making.

Standard analytical methods in business and policy—linear regression, principal component analysis (PCA), and linear programming—rely on the tacit assumption that the underlying environment is Euclidean. In practice, however, business and policy environments exhibit intrinsic nonlinearity: an identical action yields qualitatively distinct outcomes depending on the agent's position within the state space. GI theory formalises this nonlinear structure as curvature of a Riemannian manifold and brings to bear the differential-geometric apparatus pioneered in Einstein's general relativity—covariant derivatives, curvature tensors, Lie derivatives, and geodesics—upon socioeconomic data.

The paper comprises two parts. Part I lays the mathematical foundations spanning topological spaces, differentiable manifolds, tangent spaces with Riemannian metrics, connections with covariant derivatives and curvature tensors, and Lie derivatives with Killing fields; introduces the geometric AI toolkit encompassing VAEs with smooth decoders, Vector Diffusion Maps (VDM), and Neural ODEs coupled with automatic differentiation for tensor computation; and presents a complete 10-step pipeline—from data collection through manifold construction, metric learning, curvature computation, Lie derivative analysis, and optimal control simulation to visualisation and decision-making—demonstrated via two case studies (a mid-cap manufacturing firm and a national security council). Part II erects five pillars of reliability (MC Dropout for uncertainty quantification, SHAP for explainability, Double Machine Learning for causal inference, ZKML for cryptographic verification, and formal verification via Coq, Agda, Lean 4, and Z3); architects an AI Agent system operating on a digital-twin manifold; and systematises seven geometric extensions ranging from Morse theory to Sheaf theory.

The central result (Proposition 2.1) establishes that the image of a VAE decoder satisfying four conditions—domain compactness, smoothness, full-rank Jacobian, and injectivity—constitutes an embedded submanifold endowed with a well-defined Riemannian structure that is independent of both industry and data distribution. We further identify a four-layer structural gap that accounts for the absence of this framework in prior literature.

Keywords: Riemannian manifold, pullback metric, variational autoencoder, Lie derivative, covariant derivative, curvature tensor, geodesic, optimal control, Christoffel symbols, data-driven manifold, Killing field, Vector Diffusion Maps, Neural ODE, deep reinforcement learning, digital twin, MC Dropout, SHAP, Double Machine Learning, ZKML, formal verification, Morse theory, symplectic geometry, Sheaf theory

1 Introduction

1.1 The core problem: navigating a curved world with a flat map

For over a century, physicists and chemists have recognised that the structures of the natural world are most faithfully described as curved, higher-dimensional spaces—mathematically, as differentiable manifolds. In 1854, Bernhard Riemann founded the study of such spaces in his Habilitationsschrift "Über die Hypothesen, welche der Geometrie zu Grunde liegen," introducing the metric tensor as the device that encodes curvature at each point of a space of arbitrary dimension. Riemann's innovation was to unify the geometry of two-dimensional surfaces (planes, spheres) with that of three-, four-, and indeed n-dimensional "curved spaces" under a single mathematical language.

In 1915, Albert Einstein brought Riemannian geometry to its most celebrated application by completing the general theory of relativity. The Einstein field equation

$$R_{\mu\nu} - \tfrac{1}{2}\, g_{\mu\nu}\, R + \Lambda\, g_{\mu\nu} = \frac{8\pi G}{c^4}\, T_{\mu\nu} \tag{1}$$

relates the geometric properties of spacetime (left-hand side: the Ricci tensor $R_{\mu\nu}$, the scalar curvature R, and the metric tensor $g_{\mu\nu}$) to the distribution of matter and energy (right-hand side: the energy–momentum tensor $T_{\mu\nu}$). The equation articulates the reciprocal relationship whereby "matter curves spacetime, and the curvature of spacetime determines the motion of matter." In the 110 years since, physicists have relied upon this mathematics without interruption: GPS satellite orbit computation requires general-relativistic corrections (without which positional errors accumulate at approximately 10 km per day), and the imaging of black holes (Event Horizon Telescope, 2019) was made possible by computing the propagation of light through curved spacetime via Riemannian geometry.

Chemists, for their part, understand that conformational changes of molecules constitute motions on a high-dimensional curved surface (the conformation space). Roboticists exploit the fact that the motion of a robot is a path on the curved space formed by its joint angles. The configuration space of a six-joint robotic arm, for instance, is the six-dimensional torus $T^6 = (S^1)^6$, since each joint angle ranges over $[0, 2\pi)$. Across all these disciplines, prediction, control, and optimisation are achieved by *performing differential calculus on curved spaces*—that is, by computing how quantities change in each direction at each point of the manifold.

Yet in the domain of corporate management and national policy, data analysis continues to rest upon linear methods—regression analysis, principal component analysis, and linear optimisation. These methods share a tacit assumption: *the world is made of straight lines*. This paper argues that the assumption is structurally inadequate for business and policy environments, and proposes Geometric Intelligence (GI) theory as a rigorous alternative.

1.2 Why business and policy environments are curved

We illustrate the intrinsic nonlinearity of real-world environments through three concrete examples.

Exchange-rate nonlinearity. When the exchange rate moves by 1%, the impact on corporate earnings is qualitatively different at 90 JPY/USD (a strong-yen regime) versus 150 JPY/USD (a weak-yen regime). In the strong-yen regime, an additional 1% appreciation may inflict devastating damage on exporters; in the weak-yen regime, the marginal impact of a further 1% depreciation is limited. That is, "the same 1% movement" produces entirely different outcomes depending on *where in the state space* the firm currently sits. This is a nonlinear structure that a linear regression of the form $\text{earnings} = a \times \text{FX} + b$ cannot capture.

Interest-rate nonlinearity. A 25-basis-point rate increase elicits a qualitatively different market response at the zero lower bound (where it may trigger a regime change) compared with a 5% interest-rate environment (where a rise to 5.25% is a marginal adjustment).

Alliance-policy nonlinearity. The effect of strengthening an alliance on the security environment depends on the prevailing level of international stability. During periods of calm, alliance strengthening functions as "insurance"; during periods of tension, it may be perceived as "provocation," potentially intensifying the very instability it was meant to counter.

Each of these examples exhibits the defining signature of a curved space in Riemannian geometry: *the same action at different positions in the state space yields qualitatively different outcomes*. The analogy is precise: just as the Mercator projection makes Greenland appear larger than Africa (whose true area is approximately 14 times greater), and just as the shortest flight path from Tokyo to New York passes through the Arctic (not across the Pacific as a flat map would suggest), so too do flat analytical methods distort the structure of business and policy environments. Navigating by a flat map wastes fuel, squanders time, and in the worst case prevents arrival at the destination altogether. Precisely the same pathology afflicts managerial and policy decisions grounded in linear analysis.

1.3 Contributions

The principal contributions of this paper are as follows.

1. **Manifold construction pipeline.** We construct data-driven Riemannian manifolds via the pullback metric $g_{ij}(z) = (J_{f_\theta}^\top J_{f_\theta})_{ij}$ of a VAE decoder and establish the four conditions guaranteeing manifold structure (Proposition 2.1).

2. **Differential-geometric analytical apparatus.** We apply scalar curvature for stability/instability mapping, Lie derivatives for quantifying the structural impact of policies, Lie brackets for analysing inter-policy interference, geodesics for optimal path computation, and Pontryagin's maximum principle for constrained optimal control—all to socioeconomic data.

3. **Five pillars of reliability.** MC Dropout for uncertainty quantification, SHAP for explainability, DML (Double Machine Learning) for causal inference, ZKML for cryptographic verification, and formal verification (Coq, Agda, Lean 4, Z3) are integrated to provide multi-layered assurance of output trustworthiness.
4. **AI Agent architecture.** We construct the management-environment manifold as a digital-twin space and operate multiple deep-RL agents upon it, analysing emergent phenomena via the non-commutativity of Lie brackets.
5. **Systematic geometric extensions.** We organise seven extensions—Morse theory, exotic manifolds with optimal transport, surgery theory, information geometry, symplectic geometry, Weyl geometry, and Sheaf theory—into a coherent programme.

1.4 Relation to prior work

GI theory sits at the intersection of four research streams.

(1) Geometry of VAE latent spaces. Arvanitidis et al. [Arvanitidis et al., 2018] introduced the pullback metric to VAE latent spaces in their paper "Latent Space Oddity," demonstrating empirically that VAE latent spaces are non-Euclidean. GI theory extends this insight by constructing the full computational chain from the pullback metric through curvature tensors, Lie derivatives, and optimal control on socioeconomic data—a programme substantially beyond the scope of the original work.

(2) Vector Diffusion Maps. Singer & Wu [Singer & Wu, 2012] construct the graph connection Laplacian from discrete data and prove a spectral convergence theorem guaranteeing convergence to the continuous-manifold connection Laplacian. GI theory employs VDM as an independent means of validating the VAE pullback metric.

(3) Neural ODE and optimal control. Chen et al. [Chen et al., 2018] parameterise the right-hand side of an ODE with a neural network and compute gradients via the adjoint method—the discrete analogue of Pontryagin's costate equation. GI theory harnesses Neural ODE for geodesic computation and optimal control on Riemannian manifolds, exploiting the mathematical correspondence between the Neural ODE adjoint equation and Pontryagin's adjoint equation.

(4) Geometric Deep Learning. Bronstein et al. [Bronstein et al., 2021] present a unifying framework for geometric deep learning encompassing grids, groups, graphs, geodesics, and gauges. Their scope, however, does not extend to the construction of differentiable manifolds from socioeconomic data or to differential-geometric analysis for decision-making. GI theory may be viewed as an extension of the geometric deep learning paradigm into this domain.

Four-layer structural gap. We identify why this framework has not previously appeared: (1) data sparsity and educational gaps in the social sciences (GDP statistics are quarterly; differential geometry is absent from economics and business curricula); (2) the known-manifold assumption in engineering (robotics works with $SO(3)$, $SE(3)$, etc., whose geometry is known *a priori*; "learning unknown geometry from data" lies outside the standard problem formulation); (3) the two-manifold confusion in machine learning (the NPG/TRPO/PPO lineage correctly addresses the geometry of the policy parameter manifold $\mathcal{M}_{\text{weights}}$ but entirely ignores the environment state manifold $\mathcal{M}_{\text{env}}$); (4) the disciplinary intersection problem (GI theory simultaneously demands expertise in differential geometry, VAEs and Neural ODEs, deep reinforcement learning, and domain-specific business data—a combination that remains exceedingly rare as of March 2026).

2 Mathematical Foundations: From Topological Spaces to Differentiable Manifolds

2.1 Topological spaces and continuous maps

Definition 2.1 (Topological space)**.** *A pair* $(X, \mathcal{O})$ *is a* topological space *if* $\mathcal{O} \subset 2^X$ *satisfies:* (T1) $\emptyset, X \in \mathcal{O}$*;* (T2) *arbitrary unions of members of* $\mathcal{O}$ *lie in* $\mathcal{O}$*;* (T3) *finite intersections of members of* $\mathcal{O}$ *lie in* $\mathcal{O}$*. Elements of* $\mathcal{O}$ *are termed* open sets.

Definition 2.2 (Continuous map)**.** *A map* $f\colon (X, \mathcal{O}_X) \to (Y, \mathcal{O}_Y)$ *between topological spaces is* continuous *if for every open set* $V \in \mathcal{O}_Y$ *the preimage* $f^{-1}(V)$ *belongs to* $\mathcal{O}_X$*.*

Definition 2.3 (Homeomorphism)**.** *A map $f\colon X \to Y$ is a* homeomorphism *if it is bijective and both f and f^{-1} are continuous. When a homeomorphism exists, X and Y are said to be* homeomorphic.

Definition 2.4 (Compactness)**.** *A topological space X is* compact *if every open cover admits a finite subcover. By the Heine–Borel theorem, a subset of $\mathbb{R}^n$ is compact if and only if it is closed and bounded.*

Definition 2.5 (Hausdorff property)**.** *A topological space X is* Hausdorff *if for any two distinct points $p, q \in X$ there exist disjoint open sets $U \ni p$ and $V \ni q$.*

2.2 Differentiable manifolds

Definition 2.6 (Chart and atlas)**.** *A d-dimensional* chart *on a topological space M is a pair (U, φ) with $U \subset M$ open and $\varphi\colon U \to \varphi(U) \subset \mathbb{R}^d$ a homeomorphism. A collection $\mathcal{A} = \{(U_\alpha, \varphi_\alpha)\}_{\alpha \in A}$ covering M is a C^k* atlas *if every transition map $\varphi_\beta \circ \varphi_\alpha^{-1}\colon \varphi_\alpha(U_\alpha \cap U_\beta) \to \varphi_\beta(U_\alpha \cap U_\beta)$ is C^k.*

Definition 2.7 (C^k differentiable manifold)**.** *A second-countable Hausdorff space M equipped with a maximal C^k atlas is a d-*dimensional C^k differentiable manifold.

Smoothness requirement: $k \geq 3$. Throughout this paper we impose $k \geq 3$. The necessity arises from the following "smoothness chain." Computing the Riemann curvature tensor $R^l{}_{ijk}$ requires first-order partial derivatives of the Christoffel symbols Γ^k_{ij}, which are themselves constructed from first-order partial derivatives of the metric g_{ij}. Thus the curvature tensor demands second-order partial derivatives of g_{ij}, whence $g_{ij} \in C^2$. For the pullback metric, $f_\theta \in C^k$ implies $g_{ij} \in C^{k-1}$, so $g_{ij} \in C^2$ requires $k - 1 \geq 2$, i.e. $k \geq 3$. Employing C^∞ activation functions ($\tanh$, GELU, Softplus) throughout the decoder guarantees this condition with ample margin. ReLU $= \max(0, x)$, being $C^0 \setminus C^1$ at $x = 0$, fails even $k \geq 1$ and is therefore excluded.

2.3 Immersions and embeddings: conditions under which a VAE output constitutes a manifold

Definition 2.8 (Immersion)**.** *A C^k map $f\colon M \to N$ is an* immersion *if the differential $df_p\colon T_pM \to T_{f(p)}N$ is injective at every $p \in M$. When $\dim M = d$ and $\dim N = n$, this is equivalent to demanding $\operatorname{rank} J_f(p) = d$ for all p, where $J_f(p) \in \mathbb{R}^{n \times d}$ is the Jacobian.*

Definition 2.9 (Embedding)**.** *An immersion $f\colon M \to N$ is an* embedding *if it is injective and $f\colon M \to f(M)$ (with the subspace topology on $f(M)$) is a homeomorphism.*

Theorem 2.1 (Compact injective immersion is an embedding; Lee [Lee, 2012, Thm. 4.25])**.** *Let M be a compact Hausdorff space and N a Hausdorff space. Then every continuous injective immersion $f\colon M \to N$ is an embedding.*

Proof sketch. We must show that $f^{-1}\colon f(M) \to M$ is continuous. Let $C \subset M$ be closed. Since M is compact and C is closed, C is compact. The continuous image of a compact set is compact, so $f(C)$ is compact. As N is Hausdorff, compact subsets are closed; hence $f(C)$ is closed in N and *a fortiori* in $f(M)$. Thus f maps closed sets to closed sets, which is equivalent to continuity of f^{-1}. □

2.4 Manifold structure of the VAE decoder image

Proposition 2.1 (VAE decoder manifold structure — the foundational proposition of GI theory)**.** *Let $f_\theta\colon U \to \mathbb{R}^n$ be a VAE decoder satisfying:*

(i) *$U \subset \mathbb{R}^d$ is compact (closed and bounded);*

(ii) *$f_\theta \in C^k$ with $k \geq 3$ (guaranteed by C^∞ activations; ReLU excluded);*

(iii) *$\operatorname{rank} J_{f_\theta}(z) = d$ for all $z \in U$ (verified numerically via SVD: $\sigma_{\min}(J) > 0$);*

(iv) *$f_\theta|_U$ is injective.*

Then $f_\theta(U)$ is a d-dimensional C^k embedded submanifold of $\mathbb{R}^n$, and

$$g_{ij}(z) \;=\; \sum_{a=1}^{n} \frac{\partial f_\theta^a}{\partial z^i}\,\frac{\partial f_\theta^a}{\partial z^j} \;=\; \left(J_{f_\theta}^\top J_{f_\theta}\right)_{ij} \tag{2}$$

is a Riemannian metric on U.

Proof. **Step 1 (Immersion).** Since $f_\theta \in C^k$ and $\operatorname{rank} J_{f_\theta}(z) = d$ at every $z \in U$, the map $f_\theta|_U$ is a C^k immersion (Definition 2.8).

Step 2 (Promotion to embedding). U is compact (hypothesis (i)); $\mathbb{R}^n$ is Hausdorff; and $f_\theta|_U$ is a continuous, injective immersion. By Theorem 2.1, $f_\theta|_U$ is a C^k embedding, whence $f_\theta(U)$ is a C^k embedded submanifold (Lee [Lee, 2012, Prop. 5.2]).

Step 3 (Positive definiteness of the pullback metric). For any $0 \neq v \in \mathbb{R}^d$, the full-rank condition ensures $J_{f_\theta}\, v \neq 0$, so

$$v^\top g(z)\, v \;=\; \|J_{f_\theta}(z)\, v\|^2 \;>\; 0.$$

Symmetry follows from $(J^\top J)_{ij} = (J^\top J)_{ji}$. □

Failure modes when the hypotheses are violated. (i) U non-compact $\Rightarrow$ the promotion from immersion to embedding is no longer guaranteed. (ii) ReLU activations $\Rightarrow$ the immersion condition is undefined at fold hyperplanes; the second-order derivatives required for curvature computation do not exist. (iii) Rank deficiency $\Rightarrow$ g_{ij} acquires a zero eigenvalue, g^{ij} ceases to exist, and Christoffel symbols become undefined. (iv) Non-injectivity $\Rightarrow$ the image self-intersects and is not a manifold.

Remark 2.1 (Industry independence)**.** *The positive definiteness of g_{ij} follows entirely from the algebraic structure $J^\top J$ and is independent of the industry, data distribution, or statistical properties of the input. This is the mathematical fact guaranteeing Riemannian structure uniformly across all GI theory applications—manufacturing, finance, energy, national security.*

3 Tangent Spaces and the Riemannian Metric

3.1 Tangent vectors and tangent spaces

Definition 3.1 (Tangent vector)**.** *Let M be a d-dimensional differentiable manifold, $p \in M$, and $\gamma\colon (-\varepsilon, \varepsilon) \to M$ a smooth curve with $\gamma(0) = p$. Given a chart (U, φ) containing p, the* tangent vector *to γ at p is*

$$v \;=\; \dot\gamma(0) \;=\; \frac{d}{dt}\bigg|_{t=0} (\varphi \circ \gamma)(t) \;\in\; \mathbb{R}^d.$$

The set of all tangent vectors at p forms a d-dimensional vector space T_pM, the tangent space *at p.*

Tangent vectors are coordinate-independent: under a change of chart (U', φ'), the tangent vector v' is related to v by the Jacobian of the transition map. This transformation rule underpins the coordinate-transformation laws for tensor fields.

3.2 Riemannian metric and pullback metric

Definition 3.2 (Riemannian metric)**.** *A* Riemannian metric *on a differentiable manifold M is a smoothly varying family of positive-definite symmetric bilinear forms $g_p\colon T_pM \times T_pM \to \mathbb{R}$. In local coordinates, $ds^2 = g_{ij}(z)\, dz^i\, dz^j$. The Riemannian metric may be viewed as a generalisation of the Pythagorean theorem: when $g_{ij} = \delta_{ij}$ one recovers $ds^2 = (dz^1)^2 + \cdots + (dz^d)^2$.*

Definition 3.3 (Pullback metric)**.** *For a VAE decoder $f_\theta\colon U \subset \mathbb{R}^d \to \mathbb{R}^n$ satisfying Proposition 2.1, the* pullback metric *is*

$$g_{ij}(z) \;=\; \big(J_{f_\theta}(z)^\top\, J_{f_\theta}(z)\big)_{ij}, \qquad J_{f_\theta}(z) = \Big(\frac{\partial f_\theta^a}{\partial z^i}\Big) \in \mathbb{R}^{n\times d}.$$

The eigenvalues of $g_{ij}(z)$ quantify directional sensitivity: directions associated with large eigenvalues are those in which small latent-space displacements induce large data-space changes.

The physical interpretation of the pullback metric is as follows: $g_{ij}(z)$ measures, at each point and in each direction, how large a change in data space results from an infinitesimal displacement in latent space. This is quantitative information that neither UMAP nor t-SNE scatter plots can provide. The concept of the pullback metric in the VAE context was first studied empirically by Arvanitidis et al. [Arvanitidis et al., 2018].

3.3 Curve length and geodesics

Definition 3.4 (Curve length)**.** *The* length *of a smooth curve* $\gamma\colon [a,b] \to M$ *on a Riemannian manifold* (M,g) *is*

$$L[\gamma] \;=\; \int_a^b \sqrt{g_{ij}\big(\gamma(t)\big)\,\dot\gamma^i(t)\,\dot\gamma^j(t)}\;dt. \tag{3}$$

Definition 3.5 (Energy functional)**.** *The* energy *of the same curve is*

$$E[\gamma] \;=\; \tfrac{1}{2}\int_a^b g_{ij}\big(\gamma(t)\big)\,\dot\gamma^i(t)\,\dot\gamma^j(t)\;dt. \tag{4}$$

Critical points of E *with fixed endpoints satisfy the* geodesic equation

$$\ddot\gamma^k + \Gamma^k_{ij}\,\dot\gamma^i\,\dot\gamma^j = 0, \tag{5}$$

where Γ^k_{ij} *are the Christoffel symbols of the Levi-Civita connection (defined in Section 4).*

4 Connection, Covariant Derivative, and Curvature Tensor

4.1 The Levi-Civita connection and Christoffel symbols

On a curved space, the tangent spaces at distinct points are distinct vector spaces; consequently, the ordinary partial derivative does not yield the "true rate of change" of a vector field. A connection resolves this by providing a principled means of comparing vectors at different points.

Theorem 4.1 (Uniqueness of the Levi-Civita connection)**.** *On a Riemannian manifold* (M,g) *there exists a unique affine connection* ∇ *that is*

(LC1) *torsion-free:* $\Gamma^k_{ij} = \Gamma^k_{ji}$*;*

(LC2) *metric-compatible:* $\nabla_k g_{ij} = 0$.

Its Christoffel symbols are given by

$$\Gamma^k_{ij}(z) \;=\; \tfrac{1}{2}\,g^{kl}(z)\left(\frac{\partial g_{jl}}{\partial z^i} + \frac{\partial g_{il}}{\partial z^j} - \frac{\partial g_{ij}}{\partial z^l}\right). \tag{6}$$

The *covariant derivative* of a vector field W^k along ∂_i is

$$\nabla_i W^k \;=\; \frac{\partial W^k}{\partial z^i} + \Gamma^k_{ij}\,W^j. \tag{7}$$

The first term captures the component-wise rate of change; the second is a correction term accounting for the curvature of the coordinate system.

4.2 The Riemann curvature tensor

Definition 4.1 (Riemann curvature tensor)**.**

$$R^l{}_{ijk} \;=\; \partial_j\Gamma^l_{ik} - \partial_i\Gamma^l_{jk} + \Gamma^l_{jm}\,\Gamma^m_{ik} - \Gamma^l_{im}\,\Gamma^m_{jk}. \tag{8}$$

The Ricci tensor *is* $\mathrm{Ric}_{ij} = R^l{}_{ilj}$*, and the* scalar curvature *is* $\mathrm{Scal} = g^{ij}\,\mathrm{Ric}_{ij}$.

Interpretation in the management-environment context. At a point z where $\mathrm{Scal}(z) > 0$, geodesic balls have volume smaller than their Euclidean counterparts (geodesics converge; stable structure). Where $\mathrm{Scal}(z) < 0$, volumes exceed the Euclidean baseline (geodesics diverge; unstable structure). This information is *inaccessible in principle* to PCA, which projects onto linear subspaces and within whose framework curvature—an inherently nonlinear quantity—cannot even be defined.

4.3 Full-chain computation via automatic differentiation

By means of PyTorch's `autograd` module with `create_graph=True`, the following full-chain computation is executed without manual calculation: (1) Jacobian $J_{f_\theta} \to$ pullback metric g_{ij}; (2) $\partial_i g_{jl} \to$ Christoffel symbols Γ^k_{ij} via Eq. (6); (3) $\partial_j\Gamma^l_{ik} \to$ curvature tensor $R^l{}_{ijk}$ via Eq. (8); (4) vector-field derivatives $\to$ Lie brackets and Lie derivatives. Step (3) involves differentiating quantities that are themselves first-order derivatives, necessitating the `create_graph=True` flag. This is precisely why ReLU—whose "fold" at $x = 0$ renders second-order derivatives undefined—is inadmissible, whereas $\tanh$ and other C^∞ activations are required.

5 Lie Derivatives and Killing Fields

Definition 5.1 (Lie derivative of the metric)**.** *For a vector field V, the* Lie derivative *of the metric tensor g along V is*

$$(\mathcal{L}_V g)_{ij} \;=\; \nabla_i V_j + \nabla_j V_i. \tag{9}$$

When $\mathcal{L}_V g = 0$ (the Killing equation*), V is a* Killing field*, generating isometries of (M, g).*

Within GI theory, policies and managerial actions are modelled as vector fields. If the Lie derivative vanishes ($\mathcal{L}_V g = 0$, the Killing equation), the action *preserves the structure of the environment* while translating the state—a gentle intervention. If the Lie derivative is non-vanishing, the action *deforms the structure itself*: it alters which states are nearby, which are distant, which are stable, and which are unstable. This is qualitatively different from conventional policy evaluation (difference-in-differences, regression discontinuity), which measures *outcome differences* rather than *geometric deformation of the environment*.

Definition 5.2 (Lie bracket)**.** *For two vector fields V_A and V_B:*

$$[V_A, V_B]^k \;=\; V_A^i\, \partial_i V_B^k - V_B^i\, \partial_i V_A^k. \tag{10}$$

The Lie bracket quantifies the extent to which executing two policies in different orders yields different outcomes. If $[V_A, V_B] = 0$, the policies commute (order-independent). If $[V_A, V_B] \neq 0$, the outcome is order-dependent. The magnitude $\|[V_A, V_B]\|$ measures interference strength; the direction indicates the interference vector.

6 The Geometric AI Toolkit

6.1 Variational autoencoder with smooth decoder

The decoder $f_\theta : \mathbb{R}^d \to \mathbb{R}^n$ of a VAE [Kingma & Welling, 2014] is a continuous function that generates a data-space point for *any* latent-space point, smoothly interpolating between training examples. This is the decisive difference from UMAP and t-SNE, which produce *discrete point configurations* on which (i) no Riemannian metric is defined, (ii) no derivatives exist, (iii) no tensor fields can be defined, and (iv) no control-system formulation is possible. A VAE constructs a smooth manifold; UMAP and t-SNE yield scatter plots.

For the VAE output to constitute a mathematically rigorous manifold, the decoder must satisfy the conditions of Proposition 2.1. Ordinary VAEs do not guarantee these conditions. GI theory *deliberately engineers* them into the architecture and *verifies* them post-training.

6.2 Vector Diffusion Maps

Singer & Wu [Singer & Wu, 2012] construct the graph connection Laplacian from discrete data:

$$(S_\varepsilon\, v)_i \;=\; \frac{1}{d_i} \sum_j w_{ij}\, (v_i - O_{ij}\, v_j), \qquad w_{ij} = \exp\!\big(-\|z_i - z_j\|^2/\varepsilon\big), \quad d_i = \textstyle\sum_j w_{ij}, \tag{11}$$

where $O_{ij} \in O(d)$ are optimal rotation matrices between local tangent spaces.

Theorem 6.1 (Spectral convergence; Singer & Wu, 2012)**.** *Under Assumption A1, as $N \to \infty$ and $\varepsilon \to 0$ subject to the rate condition $N\varepsilon^{d/2+1} \to \infty$, the operator S_ε converges spectrally to the connection Laplacian on the tangent bundle of the underlying manifold M_0.*

6.3 Neural ODE and optimal control

Neural ODE [Chen et al., 2018] parameterises the right-hand side of an ODE: $\dot{z} = f_\theta(z, t)$, $z(0) = z_0$. Gradients are computed via the *adjoint method*:

$$\frac{da}{dt} = -a(t)^\top \frac{\partial f_\theta}{\partial z}\big(z(t), t\big), \tag{12}$$

where $a(t) = \partial L/\partial z(t)$ is the adjoint state—the discrete analogue of Pontryagin's costate equation.

When a control input $u(t) \in \mathbb{R}^m$ is present, the affine control system $\dot{z}^k = F^k(z) + u_\alpha(t)\, B^k_\alpha(z)$ is governed by Pontryagin's maximum principle via the Hamiltonian

$$H(z, p, u) = p_k\big(F^k + u_\alpha\, B^k_\alpha\big) - \ell(z, u), \qquad \dot{z}^k = \frac{\partial H}{\partial p_k}, \quad \dot{p}_k = -\frac{\partial H}{\partial z^k}. \tag{13}$$

Setting $u = 0$ and $F^k = 0$ recovers the geodesic equation (5).

7 The 10-Step GI Theory Pipeline

1. **Data collection.** Assemble industry-specific management or policy indicators.
2. **VAE manifold construction.** Train a VAE with a C^∞ decoder; verify the four conditions of Proposition 2.1.
3. **Pullback metric computation.** $g_{ij}(z) = (J^\top J)_{ij}$ via automatic differentiation.
4. **Geometric structure extraction.** Apply VDM for independent validation of the metric.
5. **Christoffel symbols.** Compute Γ^k_{ij} via Eq. (6) using automatic differentiation.
6. **Curvature tensor and scalar curvature.** Compute $R^l{}_{ijk}$ via Eq. (8). Map stable ($\mathrm{Scal} > 0$) and unstable ($\mathrm{Scal} < 0$) regions.
7. **Lie derivative computation.** Evaluate Eq. (9) to quantify structural impact of policies. Compute Lie brackets via Eq. (10) for inter-policy interference analysis.
8. **Geodesic computation.** Solve Eq. (5) numerically via Runge–Kutta.
9. **Optimal control simulation.** Apply Pontryagin's maximum principle (Eq. 13) under policy constraints.
10. **Visualisation and decision-making.** Render 3D terrain maps and generate reports.

8 Assumptions and Data Quality

8.1 Four assumptions

Assumption 1 (Manifold hypothesis (A1)). *The data $X = \{x_i\} \subset \mathbb{R}^n$ are noisy samples from a neighbourhood of a d-dimensional compact, connected, C^∞ submanifold M_0.*

Assumption 2 (Decoder regularity (A2)). *(a) $f_\theta \in C^k$, $k \geq 3$. (b) $\operatorname{rank} J_{f_\theta}(z) = d$ for all $z \in U$.*

Assumption 3 (Sufficient data (A3)). *$N\varepsilon^{d/2+1} \to \infty$. Rule of thumb: $N > 100 \times 2^d$.*

Assumption 4 (Numerical stability (A4)). *The condition number $\kappa\big(g(z)\big)$ is bounded. Regions with $\kappa > 10^4$ are flagged as numerically unreliable.*

8.2 Five data-quality checks

1. **Low-dimensional structure (A1).** PCA scree plot: $\geq 80\%$ of variance explained by a small number of components. *Case Study A:* 30 indicators of Company X; 6 principal components explain 87% → pass. *Case Study B:* 193 countries × 40 indicators; 8 components explain 82% → pass.
2. **Data density (A3).** $N > 100 \times 2^d$. *Case Study A:* $d = 6$, threshold 6,400; monthly data over 10 years yields 6,000 → marginal. *Case Study B:* $d = 8$, threshold 25,600; annual data gives 5,790 → insufficient; supplementing with monthly data raises the total to 70,000.
3. **Decoder fidelity (A2).** Reconstruction error $< 10\%$; minimum singular value of the Jacobian $> 10^{-4}$.
4. **Stability of results (A4).** Robustness to random-seed changes, bandwidth ε variation, and bootstrap resampling.
5. **Consistency with other methods.** Curvature patterns should correspond to historically known events.

9 Five Pillars of Reliability

9.1 Pillar 1: MC Dropout for uncertainty quantification

MC Dropout [Gal & Ghahramani, 2016] retains dropout at inference time, executing T forward passes on identical input to obtain predictions $\hat{y}_1, \ldots, \hat{y}_T$. The mean $\bar{y} = \frac{1}{T}\sum_t \hat{y}_t$ serves as the point estimate; the variance $\sigma^2 = \frac{1}{T}\sum_t (\hat{y}_t - \bar{y})^2$ quantifies epistemic uncertainty. Gal & Ghahramani demonstrated that this procedure corresponds to variational inference in a Bayesian neural network.

Before (without MC Dropout): the model outputs "next-quarter revenue is ¥12.0 bn" with no indication of confidence. *After*: the model outputs "next-quarter revenue is ¥12.0 bn (95% CI: ¥10.5–13.5 bn); however, the emerging-market segment has a wide confidence interval of ¥8.0–16.0 bn, indicating low predictive certainty."

9.2 Pillar 2: SHAP for explainability

SHAP (SHapley Additive exPlanations; Lundberg & Lee 2017), grounded in Shapley values [Shapley, 1953], decomposes model predictions into per-feature contributions, thereby explaining "why" a region of the manifold exhibits negative scalar curvature. Spatial SHAP on the manifold reveals how factor importance varies with position.

9.3 Pillar 3: DML for causal inference

Double Machine Learning [Chernozhukov et al., 2018] separates confounding from treatment effects via cross-fitting, enabling policy recommendations founded on *causal* rather than merely *correlational* evidence. Where SHAP answers "what is important?" (correlation), DML answers "what causes what?" (causation).

Case Study A: priority reversal. SHAP ranks "advertising expenditure increase" first, yet DML reveals that "R&D investment expansion" has the largest *causal* effect—the discrepancy arising because seasonality acts as a confounder (advertising spend and revenue both rise during peak periods). Relying on SHAP alone would have led to the recommendation "increase advertising"; incorporating DML reverses the priority to "invest in R&D."

9.4 Pillar 4: ZKML for cryptographic verification

ZKML applies zero-knowledge proofs (Goldwasser, Micali & Rackoff, 1985) to machine-learning computations: the prover generates a succinct proof (constant-size, $O(1)$ bytes) that the agreed-upon model architecture was executed correctly, verifiable without access to data or parameters. The three properties are: (i) *completeness*—correct computation always passes verification; (ii) *soundness*—fraudulent computation is always detected; (iii) *zero-knowledge*—the verifier learns nothing beyond the correctness of the computation.

Limitation. ZKML guarantees *computational correctness* (no tampering with the calculation) but does not guarantee *parameter legitimacy* (that the parameters arose from a bona fide training process). Three defence layers are therefore required: (1) ZKML for computational integrity; (2) training-log commitment via cryptographic hash chains for parameter provenance; (3) third-party audit for data-selection fairness.

9.5 Pillar 5: formal verification for safety assurance

Theorem provers (Coq, Agda, Lean 4) and the SMT solver Z3 are employed to formally verify the logical consistency of the entire pipeline.

10 Digital Twin $\times$ AI Agent Architecture

The management-environment manifold is instantiated as a server-side digital twin. Fourteen AI Agents, trained via deep reinforcement learning, operate on this manifold, executing geodesic-following movements, computing Lie derivatives of proposed interventions, and running optimal-control simulations.

Emergent phenomena—whole-system behaviours that cannot be explained by summing individual actions—are analysed through the non-commutativity of Lie brackets: $[V_A, V_B] \neq 0$ implies that the outcome depends on the execution order, giving rise to unpredictable interaction effects. A three-level monitoring system safeguards against runaway behaviour: (1) parameter-range constraints; (2) curvature-based anomaly detection; (3) formal verification of logical consistency.

11 Geometric Extensions

Morse theory: classifying and predicting tipping points. The scalar curvature function is treated as a Morse function; its critical points (maxima, minima, saddle points) are classified accordingly. Saddle points correspond to tipping points (regime changes) in the management environment; Morse inequalities provide lower bounds on their number.

Exotic manifolds and optimal transport. The existence of manifolds that are homeomorphic but not diffeomorphic (exotic manifolds) implies that management environments with identical coarse-grained structure may nonetheless differ in their smooth structure. Combined with optimal transport theory, this observation opens pathways to early detection of large-scale structural shifts such as pandemics.

Surgery theory: modelling radical restructuring. Surgery theory—the mathematical operation of modifying a manifold's topology in a controlled manner—provides a model for radical organisational restructuring (mergers and acquisitions, regime change).

Symplectic geometry: long-horizon simulation stability. Symplectic integrators, which preserve the Hamiltonian structure of the dynamics, guarantee energy conservation and numerical stability over long simulation horizons.

Weyl geometry: gauge invariance. Weyl-geometric gauge invariance ensures that analytical results are independent of the choice of currency denomination (USD, EUR, JPY). The *sign* of the scalar curvature is gauge-invariant, mathematically guaranteeing that "the same conclusion obtains regardless of the currency in which the computation is performed."

Sheaf theory: local consistency verification. The Sheaf Laplacian detects local inconsistencies in data or model outputs across overlapping regions of the manifold, thereby flagging data tampering or the violation of modelling assumptions.

12 Case Studies

12.1 Case Study A: management-environment manifold of manufacturing Company X

Company X is a mid-cap manufacturer with approximately ¥100 bn in annual revenue. Its core domestic market has matured, and it is considering expansion into emerging markets.

Dataset. Proprietary data (50 outlets $\times$ monthly; revenue, average transaction value, footfall, inventory turnover, headcount, customer satisfaction, etc.—15 indicators $\times$ 10 years), macroeconomic data (GDP growth, inflation, unemployment, consumer confidence index, exchange rate, interest rate, commodity price index), industry data (market size, competitor share, patent filings), and emerging-market data (target-country GDP, demographics, middle-class ratio, political stability index, regulatory environment index). In total: approximately 30 indicators and 6,000 data points. A 6-dimensional management-environment manifold is constructed.

Four questions and their geometric answers. **Q1** (distance): Compute the Riemannian distance between the domestic-market and emerging-market states; the discrepancy from the Euclidean distance reflects the true difficulty of transition. **Q2** (curvature): Identify negative-scalar-curvature regions (structurally unstable zones) along the transition path. **Q3** (optimal path): Compute the geodesic and the Pontryagin-optimal path under policy constraints. **Q4** (Lie bracket): Quantify interference among three candidate policies—cost reduction, market expansion, and technology investment.

Verification of Proposition 2.1. (i) Region of interest $\|z\| \leq 3$: compact ✓. (ii) $\tanh$ activations: C^∞ ✓. (iii) SVD at 100 representative points: $\sigma_{\min} = 8.3 \times 10^{-3} > 10^{-4}$, full rank ✓. (iv) Approximate injectivity check on 1,000 random pairs: no collisions ✓.

12.2 Case Study B: international-affairs manifold of a national security council

193 countries $\times$ 40 indicators (military: defence spending/GDP ratio, nuclear forces, conventional forces, cyber capability, space capability; diplomatic: alliance network, UN General Assembly voting patterns, bilateral treaty counts, summit frequency; economic security: bilateral trade dependency matrix, energy import concentration, strategic-material supply concentration, USD-transaction dependence; socio-political: democracy index, press freedom, regime stability index) $\times$ 30 years. An 8-dimensional international-affairs manifold is constructed.

Data-sparse regions. For certain authoritarian states with limited data disclosure, $\sigma_{\min}$ drops to 3.2×10^{-5}—below the 10^{-4} threshold. The analysis proceeds for G20 and major allied nations; medium-confidence flags are attached to data-sparse medium-sized states; and additional intelligence collection is recommended for the sparsest regions. Manifold analysis is *not* uniformly reliable across all regions; results should always be accompanied by a *confidence map*.

13 Applications and Ethics

13.1 Conflicts of interest

Simultaneously contracting with two competing firms (or two rival governments) exposes the intelligence provider to ethical risk. ZKML's three defence layers partially mitigate the technical dimension, but fairness in data selection

ultimately requires institutional safeguards (third-party audit).

13.2 An incompleteness theorem for geometric modelling

When the conditions of Proposition 2.1 or Assumptions A1–A4 are violated, results are unreliable. Explicitly delineating the boundary between what GI theory can and cannot address is itself a substantive contribution.

13.3 Information symmetry and asymmetry

We consider the game-theoretic implications of scenarios in which an adversary operates an equivalent AI system, analysing equilibria under both information symmetry (both parties employ GI theory) and information asymmetry (only one does).

14 Conclusion

We have proposed Geometric Intelligence (GI) theory: a framework that constructs data-driven Riemannian manifolds via the pullback metric of a VAE decoder and applies the complete differential-geometric toolkit—covariant derivatives, curvature tensors, Lie derivatives, geodesics, and optimal control—to corporate management and national-policy decision-making.

The central claim of this paper is that business environments are *curved, not flat*: to neglect this curvature is to incur the same category of structural error as navigating the Earth's surface by means of a flat map. GI theory corrects this error with mathematical rigour.

Five pillars of reliability (MC Dropout, SHAP, DML, ZKML, formal verification) furnish multi-layered assurance of output trustworthiness. An AI Agent architecture enables autonomous policy exploration on the manifold. Seven geometric extensions enlarge the theoretical scope.

The theory's fundamental limits are contingent upon the four conditions of Proposition 2.1 and Assumptions A1–A4; when these are not met, the reliability of results is not guaranteed. The explicit demarcation of these limits is itself a principal contribution.

Five guiding principles govern the pipeline: (1) *Purpose first*—the objective is to generate intelligence upon which one can act; (2) *Do not add; raise reliability*; (3) *Separate the known from the unknown*; (4) *Data quality is everything*—just as the quality of shortbread is determined by its three ingredients (butter, sugar, flour), so the quality of the pipeline is determined by its data; (5) *Mathematics does not bend*—the sign of the scalar curvature is the same regardless of who computes it; however, the *choice* of input data admits human judgement, and therein lies the potential for conflicts of interest.

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